

Platform Software · Optical Tracking

SKADI Tracking Software (API and Viewer)

Make the in-house SKADI optical 3D tracker directly usable by surgical-navigation companies and research groups through its software layer.

Period 2023.02 - present | Evidence state Ongoing | Portfolio tier Platform Software

Develop and maintain the SKADI API and SDK, the Viewer that shows device state and tracking results, and a 3D Slicer custom-application template that lets researchers start immediately.

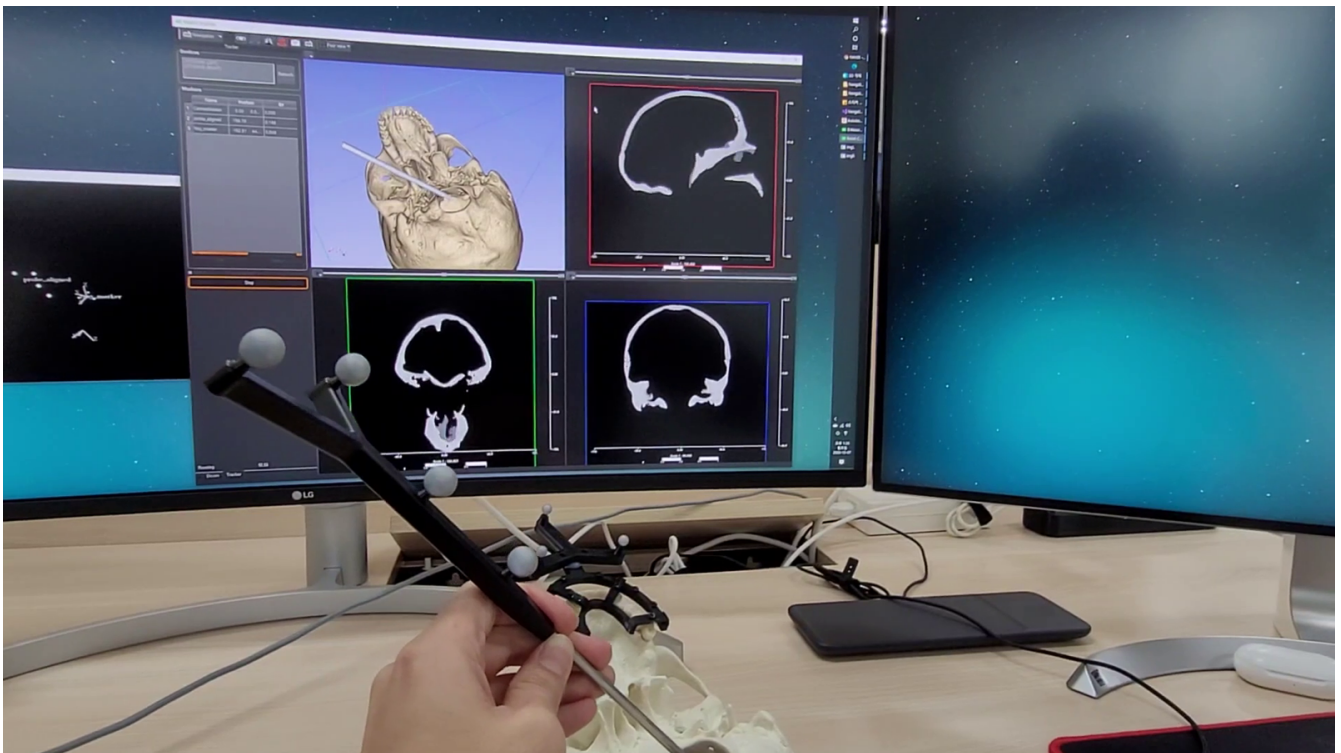


Figure 1. Demonstration clip: the SKADI Viewer shows a tracked instrument over a skull phantom in the 3D model and section views.

Problem and system boundary

An optical tracker is not used as bare hardware: coordinate frames, marker definitions, real-time streaming, and error states must be exposed through an interface application developers can work with.

Owned decisions and implementation

My role

Own the design, implementation, and maintenance of the API, SDK, and Viewer; wrote the 3D Slicer custom-application template; support customer integrations and enquiries.

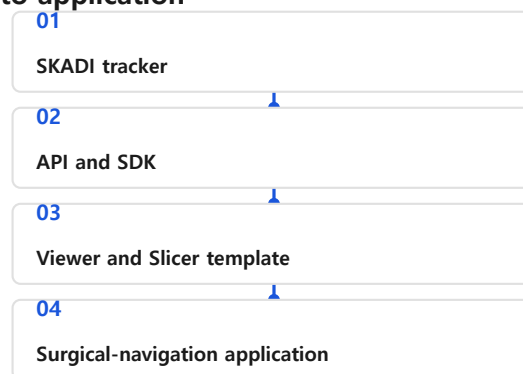
SDK layers

Device connection, marker and frame definitions, real-time streaming, and error states are exposed through the API with language bindings.

Viewer and template

The Viewer verifies device state and tracking output; the Slicer template lets researchers start a navigation prototype at once.

Software layers from tracker to application



Explanatory system relationship diagram; it is not photographic or experimental evidence.

Verification and evidence ledger

Viewer screens, an API structure diagram, and the Slicer template in action are the evidence; customer-site footage is excluded.

Working evidence

Working Viewer, API, and template screens are the evidence; customer-site footage is excluded.

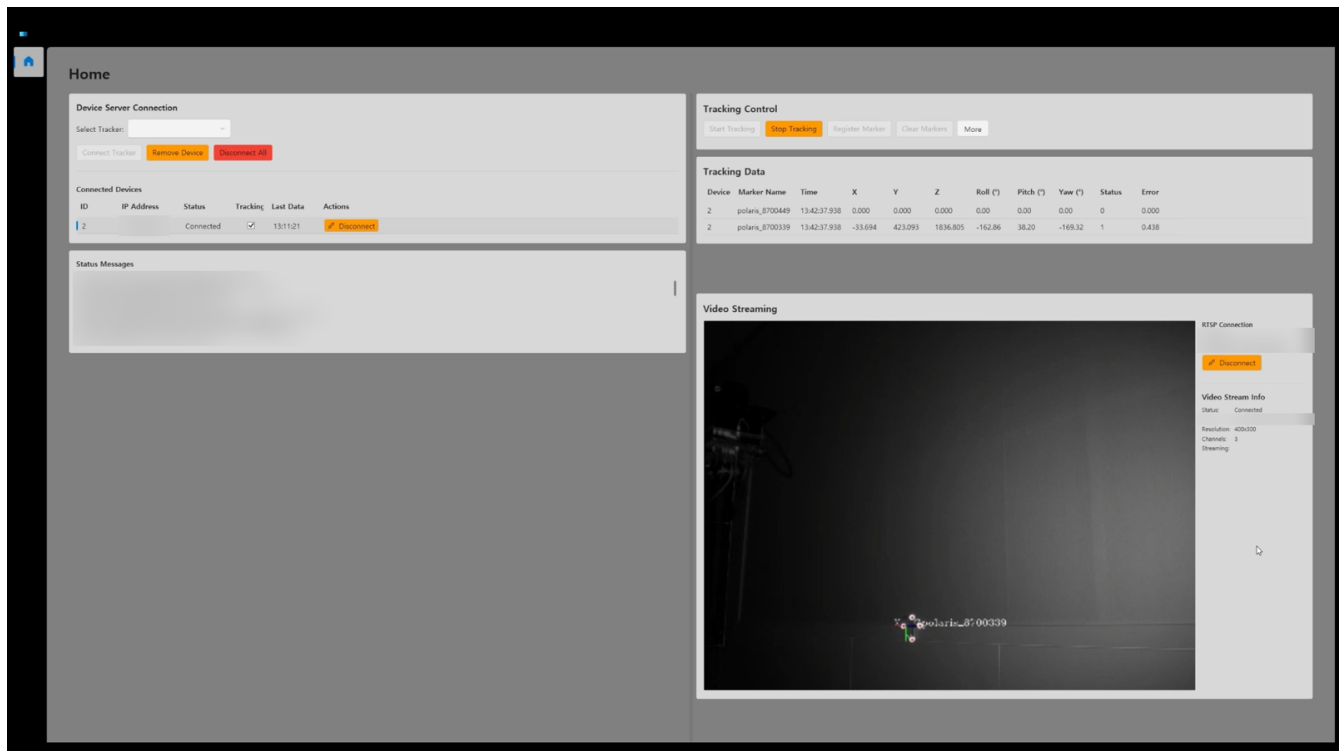


Figure 2. SKADI Viewer home: device connection, tracking-data table, and video streaming (network addresses hidden)

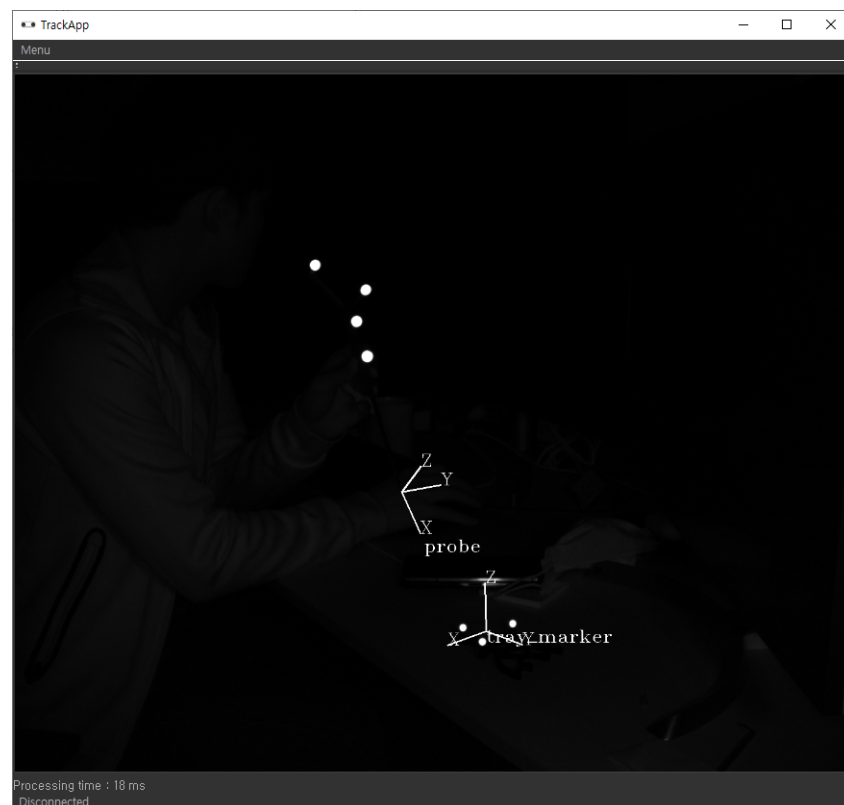


Figure 3. Tracking view: marker detection on the IR image with probe and tray-marker frames

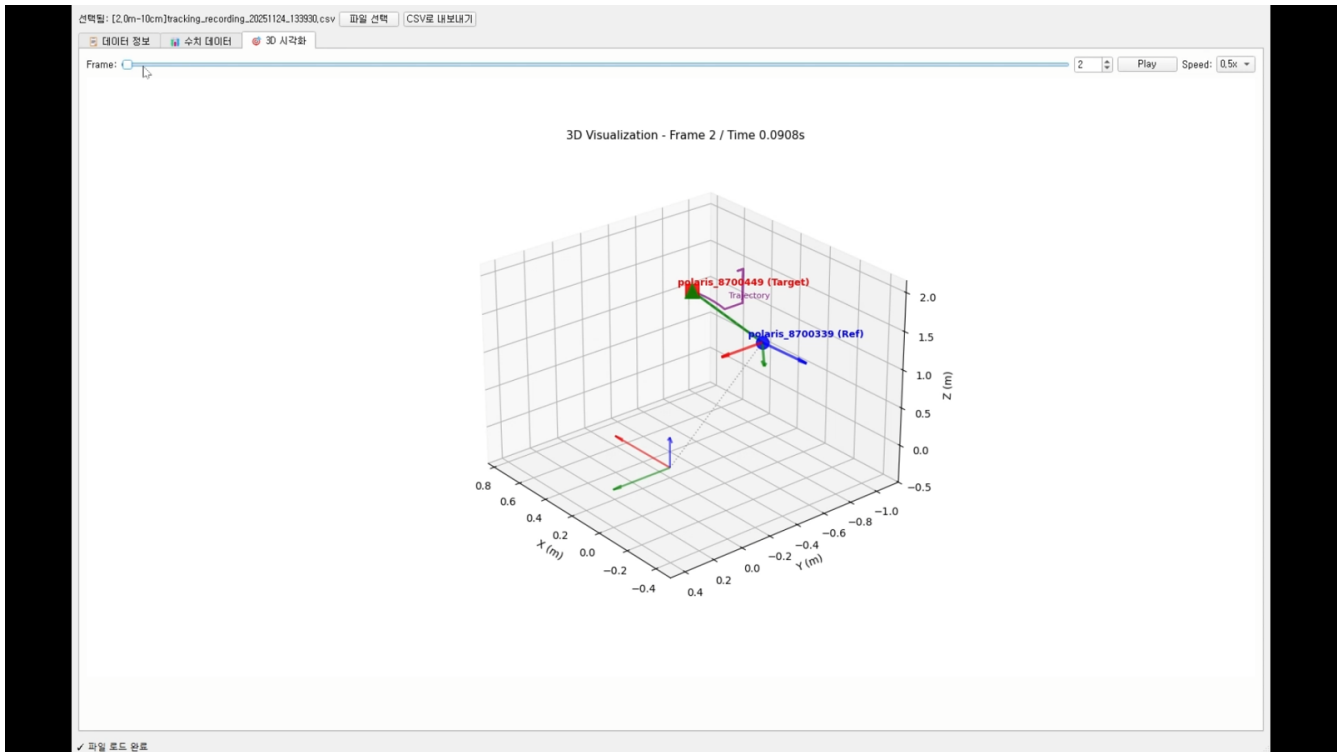


Figure 4. Recording viewer: 3D playback of recorded marker poses

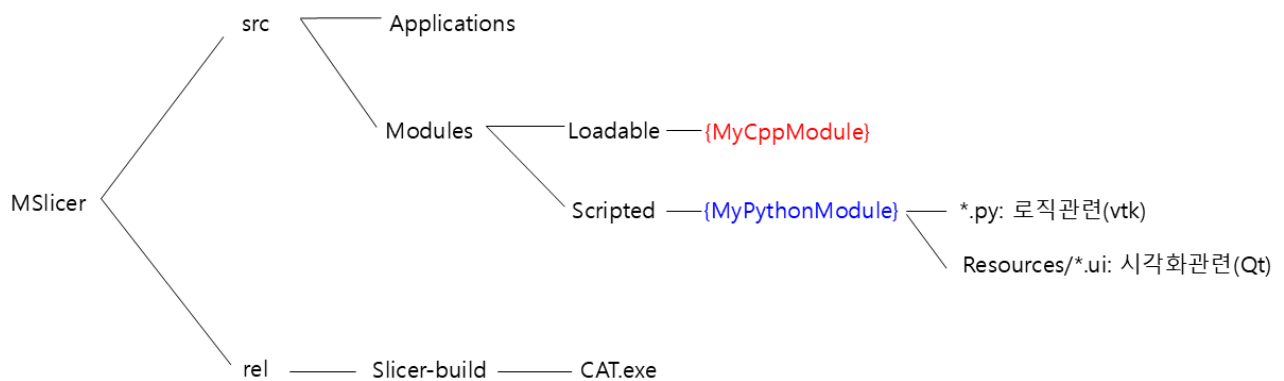


Figure 5. 3D Slicer custom-app template layout: Loadable and Scripted modules and build output



Figure 6. Granted patent certificate for "Position tracking device and method" (application 10-2024-0186864)

Role, team result, and limits

Team result

Device hardware, optical and mechanical design, sales, and delivery belong to other colleagues. Delivery records and revenue are company results and are not claimed here.

Limitations

Device specifications, accuracy figures, customer lists, and sales numbers are company-owned and not published.

Specifications, accuracy, and sales are company-owned; this case covers the software layer only.

Collaboration boundary

Interfaces are agreed with optical and hardware designers, customer navigation developers, and research users.

Implementation technologies

SKADI · C++ · Python API · Viewer · 3D Slicer · Optical tracking

Primary capability

Medical Navigation & Visualization · 3D Geometry & Registration

Registered public evidence

No external public link is currently registered.

What to bring

Share the problem, data or sensors, target validation, and schedule. We can define coordinate and evidence boundaries together.

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